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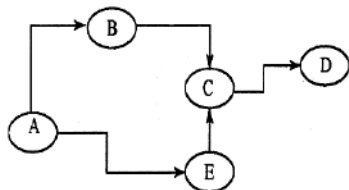
By Amit Katiyar (MCA-JNU)



22. There are 6561 balls out of them 1 is heavy. Find the minimum number of times the balls have to be weighed for finding out the heavy ball.

- (a)12 (b)9 (c)8 (d)15

Directions : Questions 23 to 25 : The following sketch shows the pipeline carrying material from one location to another. The capacity of each pipeline is 2000. The demand for the material at B is 800, at C is 800, at D is 1400 and at E is 400. The arrow indicates the direction of (material flow through pipeline. The flow through pipelines meets exactly the demand at each location. Flow from B to C is 600.



23. The quantity moved from A to E is

- (a)400 (b)1600
(c)1400 (d)2000

24. The free capacity available in the A-B pipeline is

- (a)0 (b)200
(c)400 (d)600

25. What is the free capacity available in the E – C pipeline ?

- (a)600 (b)400
(c)200 (d)0

26. A child can do a piece of work 15 hours slower than woman ? The child works, for 18 hours on the job and then the woman takes charge for 6 hours. In this manner, $\frac{3}{5}$ of the work can be completed. To complete the job now, how much time the woman take ?

- (a)24 hours (b)18 hours
(c)12 hours (d)30 hours

Directions : Questions 27 and 28 : In the following questions, select one alternative in which the third statement is implied by the first two statements.

27. (a) All elephants are wild. All lions are wild. So all lions are elephants.
(b) All mangoes are red. Some apples are mangoes. So all apples are red
(c) All roads are boxes. All foxes are roads. So all boxes are foxes.
(d) All X Y Z can run. All ABC are XYZ. So all ABC can run.

28. (a) All dogs are mad. All sick persons are mad. So all sick persons are dogs
(b) All oranges are black. All figs are oranges. So all figs are black.
(c) All windows are dogs. Some doors are dogs. So all windows are doors.
(d) No man can fly. No kite can fly. So all men are kites.

29. In the following square, numbers have been filled according to some rule. One space has been left blank. Find the correct number out of those given below for the blank space.

56	65	78
12	?	30
44	14	48

- (a)14 (b)44
(c)62 (d)51

30. A culprit was spotted by the police from a distance of 250 m. When the police men started running towards the culprit at a speed of 10 km/h, the culprit also fled. If his speed was 8 km/h, find out how far the culprit had run before he was overpowered.

- (a)2 km (b)1 km
(c)1.5 km (d)0.8 km

Directions: Questions 31 to 33: In each of the following three questions, four numbers are given. Out of these, three are alike in a certain way but the rest one is different. Choose the one which is different from the rest three .

31. (a)2384 (b)3629 (c)3756 (d)4298
32. (a)325 (b)236 (c)178 (d)639
33. (a)5698 (b)4321 (c)7963 (d)4232

34. Find the word that names a necessary part of the underlined word. Gala

- (a)Celebration (b)Tuxedo
(c)Appetizer (d)Orator

35. Which should be the next two numbers in the series 28 25 5 21 18 5 14

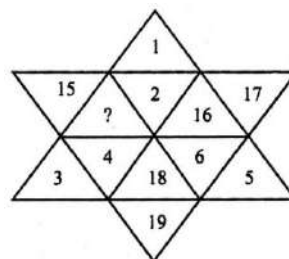
- (a)11, 5 (b)10, 7
(c)11, 8 (d) 5, 10

36. In a certain code language, '617' means 'sweet and hot'; '735' means 'coffee is sweet' and '263' means 'tea is hot'. Which of the following would mean 'coffee is hot' ?

- (a)731 (b)536
(c)367 (d)753

Directions : Questions 37 and 38 : Find the missing number in each of the following questions :

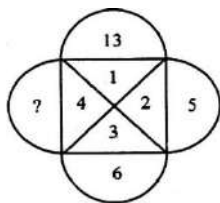
37.



- (a)13 (b)14
(c)20 (d)21



38.



- (a)10 (b)11
(c)12 (d)13

39. How many numbers from 1 to 100 are such each of which is divisible by 8 and whose at least one digit is 8

- (a)Four (b)Zero
(c)Eight (d)Six

Direction : Question 40 to 42 : Read the following information carefully and answer the question that follow :

- There is group of five person – P, Q, R, S and T
- One of them is horticulturist, one is a physicist, one is a journalist, one is an industrialist and one is an advocate.
- Three of them – P, R and advocate prefer tea to coffee and two of them – Q and the journalist prefer coffee to tea.
- The industrialist, S and P are friends to one another but two of these prefer coffee to tea
- The horticulturist is R's brother.

40. Who is a horticulturist ?

- (a)P (b)Q (c)R (d)S

41. Who is an industrialist ?

- (a)T (b)R (c)Q (d)S

42. Which of the following groups include a person who likes tea but is not an advocate ?

- (a)PRT (b)ST (c)QRT (d)None

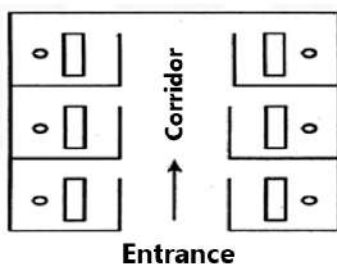
43. If the direction North-East becomes South-East, how will other directions change ?

- (a)West to North (b)South to South-West
(c)North-West to East (d)East to South-West

44. How many numbers between 1 to 1000 (both excluded) are both squares and cubes ?

- (a)none (b)1 (c)2 (d)3

Directions: Questions 45 to 48 : The plan given below, shows office for six officers namely A, B, C, D, E, and F. Both B and C occupy offices to the right of the corridor (as one enters the office block) and A occupies the office to the left of the corridor. E and F occupy offices on opposite sides of the corridor but their offices do not face each other. The offices of C and D face each other. E does not have a corner office. F's office is further down the corridor than A's, but on the same side.



45. If E sits in his office and faces the corridor, whose office is to his left ?

- (a)A (b)B (c)C (d)D

46. Whose office faces A's office ?

- (a)B (b)C (c)D (d)E

47. Who is/are F's neighbour(s) ?

- (a)A only (b)A and D (c)C only (d)B and C

48. D was heard telling someone to go further down the corridor to the last office on the right. To whose room was he trying to direct that person ?

- (a)A (b)B (c)C (d)F

49. Persons X, Y, Z and Q live in red, green, yellow or blue colored houses placed in a sequence on a street. Z lives in a yellow house. The green house is adjacent to the blue house. X does not live adjacent to Z. The yellow house is in between the green and red house. The color of the house X lives in is

- (a)Green (b)Blue (c)Red (d)none

50. Which two months in a year have the same calendar ?

- (a)June – October (b)April – November
(c)April – July (d)October – December

ENGLISH

Directions : Questions 51 and 52 : In each of the following sentences, a part of the sentence is underlined. Beneath each sentence, four different ways of phrasing the underlined part are indicated.

Choose the best alternative from among the four.

51. It was us who had left before he arrived.

- (a)We who had left before time he had arrived
(b)Us who has went before he arrived
(c)Us who had went before he had arrived
(d)We who had left before he arrived

52. Many of these environmentalists proclaim to save nothing less than the planet itself.

- (a)to save nothing lesser than
(b)that they are saving nothing less than
(c)that they save nothing less than
(d)to have saved nothing less than

53. In the question, the word "BUNDLE" is used in four different ways, numbered A to D. Choose the option in which the usage of the word is INAPPROPRIATE.

- (a) The newborn baby was a bundle of joy for the family.
(b) Mobile operators are offering a bundle of additional benefits.
(c) He made a bundle in the share market.
(d) It was sheer luck that brought a bundle of boy-scouts to where I was lying wounded.



54. Fill in the blank :

The sugar dissolved in water.....; finally all that remained was an almost residue on the bottom of the glass.

- (a) Quickly ... lumpy
- (b) Immediately ... fragrant
- (c) Gradually ... imperceptible
- (d) Subsequently ... glassy

55. There are four statements, of which one is incorrect. Choose the incorrect one.

- (a) A hater of the institution of marriage is misogynist.
- (b) The violation of sacred things is sacrilege.
- (c) To prevaricate is to make evasive or misleading statements.
- (d) A torpid person is generally hyperactive

56. The sentences given below, when properly sequenced, form a coherent paragraph. Each sentence is labeled with a letter. Choose the most logical order of sentences from among the given choices to construct a coherent paragraph.

- (P) Surrendered, or captured, combatants cannot be incarcerated in tazor wire cages; this 'war' has a dubious legality.
- (Q) How can then one characterize a conflict to be waged against a phenomenon as war ?
- (R) The phrase 'war against terror', which has passed into the common lexicon, is a huge misnomer.
- (S) Besides, war has a juridical meaning in international law, which has confided the laws of war, imbuing them with a humanitarian content.
- (T) Terror is a phenomenon, not an entity - either State or non-State.

- (a) TRSQP (b) QTRSP
- (c) TQRPS (d) RTQSP

Direction : Questions 57 and 58 : Select the pair of words which are related in the same way as the capitalized words are related to each other

57. MOTH : CLOTHING ::

- (a) egg : larva (b) hole : repair
- (c) suit : dress (d) stigma : reputation

58. ASCETIC : LUXURY ::

- (a) philosopher : knowledge
- (b) general : victory
- (c) misogynist : women
- (d) teacher : blackboard

59. Out of the four alternatives, choose the one which best expresses the meaning of the given word : CLEMENCY

- (a) Empathy (b) Kindness
- (c) Sympathy (d) Forgiveness

60. Choose the word that is opposite in meaning to the word COMPOSE

- (a) disturb (b) reveal
- (c) strengthen (d) isolate

61. Following sentence has one or two blanks, each blank indicating that something has been omitted. Beneath the sentence are given four words or sets of words. Choose the word or set of words for each blank that best fits meaning of the sentence as a whole.

His presentation was so lengthy and.....that it was difficult for us to find out the real.....in it.

- (a) verbose, content (b) tedious, skill
- (c) laborious, coverage (d) simple, meaning

62. Find the synonym that is most nearly similar in meaning to the word **CLANDESTINE**

- (a) abortive (b) secret
- (c) tangible (d) doomed

63. Choose the word opposite in meaning to the given word : **FLAMBOYANT**

- (a) Quiet (b) Excited
- (c) Disturbed (d) Distressed

Directions : Questions 64 and 65 : Read the passage and select the most suitable answer to questions from the given choices.

Observe the dilemma of the fungus : it is a plant, but it possesses no chlorophyll. While all other plants put the sun's energy to work for them combining the nutrients of ground and air into the body structure, the chlorophyllless must look elsewhere for energy supply. It finds it in those other plants which, having received their energy free from the sun, relinquish it at some point in their cycle either to animals (like us humans) or to the fungi.

In this search for energy the fungus has become the earth's major source of rot and decay %. Whereever you see mold forming on a piece of bread, or a pile of leaves turning to compost, or a blown-down tree becoming pulp on the ground, you are watching a fungus eating. Without fungus action the earth would be piled high with the dead plant life of past centuries. In fact, certain plants which contain resins that are toxic to fungi will last indefinitely; specimens of the redwood, for instance, can still be found resting on the forest floor centuries after having been blown down.

64. The passage states all the following about fungi EXCEPT :

- (a) They are responsible for the decomposition of much plant life
- (b) They cannot live completely apart from other plants
- (c) They are vastly different from other plants
- (d) They are poisonous to resin-producing plants

65. The passage is primarily concerned with

- (a) Warning people of the dangers of fungi
- (b) Rot and decay of plants in nature
- (c) Describing the action of fungi
- (d) Relating how most plants use solar energy



66. The pilot was.....injured; he died within half an hour.
 (a)seriously (b)fatally
 (c)fatefully (d)vitaly
67. A million cinemas a year bring the same stale bladerdash.
 (a) adventure (b)nonsense
 (c) thriller (d) romance
68. There are several ways of the price at which a product can be marketed.
 (a)arriving (b)thinking
 (c)determining (d)noticing
69. The police.....the mob.
 (a)scattered (b)disbanded
 (c)drove (d)dispersed
70. There must be lively discussion of Indian authors if we are to foster our knowledge.
 (a)promote (b)cherish
 (c)nourish (d)nurture

MATHEMATICS

71. The distance between the parallel lines $y = 2x + 4$ and $6x = 3y + 5$
 (a) $\frac{17}{\sqrt{3}}$ (b)1 (c) $\frac{3}{\sqrt{5}}$ (d) $\frac{17\sqrt{5}}{15}$
72. If any tangent to the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ intercepts equal length l on the axes, then $l =$
 (a) $a^2 + b^2$ (b) $\sqrt{a^2 + b^2}$
 (c) $(a^2 + b^2)^2$ (d)None
73. The rate of increase of length of the shadow of a man 2 meters height, due to a lamp at 10 meters height, when he is moving away from it at the rate of 2m/sec is
 (a) $\frac{1}{2}$ m/sec (b) $\frac{2}{5}$ m/sec
 (c) $\frac{1}{3}$ m/sec (d)5 m/sec
74. If the two pairs of lines $X^2 - 2mXY - Y^2 = 0$ and $X^2 - 2nXY - Y^2 = 0$ are such that one of the them represents the bisector of the angles between the other, then
 (a) $mn + 1 = 0$ (b) $mn - 1 = 0$
 (c) $\frac{1}{m} + \frac{1}{n} = 0$ (d) $\frac{1}{m} - \frac{1}{n} = 0$
75. If $\sin^{-1} \frac{2a}{1+a^2} - \cos^{-1} \frac{1-b^2}{1+b^2} = \tan^{-1} \frac{2x}{1-x^2}$ then x is equal to
 (a)a (b)b (c) $\frac{a+b}{1-ab}$ (d) $\frac{a-b}{1+ab}$
76. Let $\omega \neq 1$ be a cube root of unity and $i = \sqrt{-1}$. The value of the determinant $\begin{vmatrix} 1 & 1+i+\omega^2 & \omega^2 \\ 1-i & -1 & \omega^2-1 \\ -i & -i+\omega-1 & -\omega^3 \end{vmatrix}$ is
 (a)0 (b) ω (c) ω^2 (d) $1 + \omega^2$
77. If C is the middle point of AB and P is any point outside AB, then
 (a) $P\vec{A} + P\vec{B} = P\vec{C}$ (b) $P\vec{A} + P\vec{B} = 2P\vec{C}$
 (c) $P\vec{A} + P\vec{B} + P\vec{C} = \vec{O}$ (d) $P\vec{A} + P\vec{B} + 2P\vec{C} = \vec{O}$

78. If a, b, c are in A.P., p, q, r are in H.P. and ap, bq, cr are in G.P. then $\frac{p}{r} + \frac{r}{p}$ is equal to
 (a) $\frac{a}{c} - \frac{c}{a}$ (b) $\frac{a}{c} + \frac{c}{a}$ (c) $\frac{b}{q} - \frac{a}{p}$ (d) $\frac{b}{q} + \frac{a}{p}$
79. The position vector of A, B, C and D are $\hat{i} + \hat{j} + \hat{k}, 2\hat{i} + 5\hat{j}, 3\hat{i} + 2\hat{j} - 3\hat{k}$ and $\hat{i} - 6\hat{j} - \hat{k}$, then the angle between $\overrightarrow{AB}$ and $\overrightarrow{CD}$ is
 (a)0 (b) $\frac{\pi}{4}$ (c) $\frac{\pi}{2}$ (d) π
80. The value of $X^4 + 9X^3 + 35X^2 - X + 4$ for $X = -5 + 2\sqrt{-4}$ is
 (a)0 (b)-160 (c)160 (d)-164
81. A vector $\vec{a}$ has components $2p$ and 1 with respect to a rectangular Cartesian system. This system is rotated through a certain angle about the origin in the counter clockwise sense. If, with respect to the new system, $\vec{a}$ has components $p + 1$ and 1 , then
 (a) $p = 0$ (b) $p = 1$ or $p = -\frac{1}{3}$
 (c) $p = -1$ or $p = \frac{1}{3}$ (d) $p = 1$ or $p = -1$
82. The average marks per student in a class of 30 students were 45. On rechecking it was found that marks had been entered wrongly in two cases. After correction these marks were increased by 24 and 34 in the two cases. The corrected average marks per student are
 (a)75 (b)60 (c)56 (d)47
83. If $I_1 = \int_0^1 2^{x^2} dx, I_2 = \int_0^1 2^{x^3} dx, I_3 = \int_1^2 2^{x^2} dx$ and $I_4 = \int_1^2 2^{x^3} dx$ then
 (a) $I_3 = I_4$ (b) $I_3 > I_4$
 (c) $I_2 > I_1$ (d) $I_1 > I_2$
84. One hundred identical coins, each with probability p of showing up a head, are tossed. If $0 < p < 1$ and if the probability of heads on exactly 50 coins is equal to that of heads on exactly 51 coins then the value of p , is
 (a) $\frac{1}{2}$ (b) $\frac{49}{101}$ (c) $\frac{50}{101}$ (d) $\frac{51}{101}$
85. The value of the integral $\int_0^{\frac{\pi}{4}} \frac{\sin x + \cos x}{3 + \sin 2x} dx$ is
 (a) $\log 2$ (b) $\log 3$ (c) $\frac{1}{4} \log 3$ (d) $\frac{1}{8} \log 3$
86. India plays two matches each with West Indies and Australia. In any match the probabilities of India getting points 0, 1 and 2 are 0.45, 0.05 and 0.50 respectively. Assuming that the outcomes are independent, the probability of India getting at least 7 points is
 (a)0.8750 (b)0.0875
 (c)0.0625 (d)0.0250
87. If the tangents at the extremities of a focal chord of the parabola $x^2 = 4ay$ meet the tangent at the vertex at points whose abscissa are x_1 and x_2 then $x_1 x_2 =$
 (a) a^2 (b) $a^2 - 1$ (c) $a^2 + 1$ (d) $-a^2$



88. Set A has 3 elements and set B has 4 elements. The number of injection that can be defined from A to B is
(a)144 (b)12 (c)24 (d)64
89. The angle of intersection of the cardioids $r = a(1 + \cos \theta)$, $r = a(1 - \cos \theta)$ is
(a) $\pi/2$ (b)0 (c) $\pi/4$ (d) π
90. Identify the wrong statement from the following :
(a) If A and B are two sets, then $A - B = A \cap B$
(b) If A, B and C are sets, then $(A - B) - C = (\bar{A} - C) - (B - C)$
(c) If A and B are two sets, then $\overline{A \cup B} = \overline{A \cap B}$
(d) If A, B and C are sets, then $A \cap B \cap \bar{C} \subseteq A \cap B$
91. The circle $x^2 + y^2 = 9$ is contained in the circle $x^2 + y^2 - 6x - 8y + 25 = c^2$ if
(a) $c = 2$ (b) $c = 3$ (c) $c = 5$ (d) $c = 10$
92. A person stands at a point A due south of a tower and observes that its elevation is 60° . He then walks westwards towards B, where the elevation is 45° . At a point C on AB produced, he finds it to be 30° . Then AB/BC is equal to
(a)1/2 (b)1 (c)2 (d)5/2
93. The point (4, 1) undergoes the following three transformation successively :
(i) reflection about the line $y = x$
(ii) Transformation through a distance 2 unit along the positive direction of x-axis
(iii) rotation through an angle of $\pi/4$ about the origin in the anticlockwise direction. The final position of the point is given by the coordinates.
(a) $(\frac{-1}{\sqrt{2}}, \frac{7}{\sqrt{2}})$ (b) $(\frac{1}{\sqrt{2}}, \frac{7}{\sqrt{2}})$
(c) $(-2, 7\sqrt{2})$ (d) $(\sqrt{2}, 7\sqrt{2})$
94. In a triangle ABC, R is circumradius and $8R^2 = a^2 + b^2 + c^2$. The triangle ABC is
(a)Acute angled (b)Obtuse angled
(c)Right angled (d)None of these
95. If $a \neq p, b \neq q, c \neq r$ and $\begin{vmatrix} p & b & c \\ a & q & c \\ a & b & r \end{vmatrix} = 0$, then the value of $\frac{p}{p-a} + \frac{q}{q-b} + \frac{r}{r-c}$
(a)0 (b)1 (c)-1 (d)2
96. The value of $\sqrt{3} \cot 20^\circ - 4 \cos 20^\circ$ is
(a)0 (b)1 (c)-1 (d)None
97. If $y = a \log x + bx^2 + x$ has its extremum value at $x = -1$ and $x = 2$, then
(a) $a = 2, b = -1$ (b) $a = -2, b = \frac{1}{2}$
(c) $a = 2, b = -\frac{1}{2}$ (d) $a = 1, b = -\frac{1}{2}$
98. Let $\vec{a}, \vec{b}$ and $\vec{c}$ be three non zero vectors, no two of which are collinear and the vector $\vec{a} + \vec{b}$ is collinear with $\vec{c}$ while $\vec{b} + \vec{c}$ is collinear with $\vec{a}$, then $\vec{a} + \vec{b} + \vec{c}$ is equal to
(a) $\vec{a}$ (b) $\vec{b}$ (c) $\vec{c}$ (d)None
99. The value of 'a' for which the system of equations $a^3x + (a+1)^3y + (a+2)^3z = 0$
 $ax + (a+1)y + (a+2)z = 0$
 $x + y + z = 0$ has a non zero solution, is
(a) 1 (b) 0 (c) -1 (d) None
100. The vectors $\vec{a}, \vec{b}$ and $\vec{c}$ are equal in length and taken pairwise make equal angles. If $\vec{a} = \hat{i} + \hat{j}, \vec{b} = \hat{j} + \hat{k}$, and $\vec{c}$ make an obtuse angle with the base vector $\hat{i}$, then $\vec{c}$ is equal to
(a) $\hat{i} + \hat{k}$ (b) $-\hat{i} + 4\hat{j} - \hat{k}$
(c) $-\frac{1}{3}\hat{i} + \frac{4}{3}\hat{j} - \frac{1}{3}\hat{k}$ (d) $\frac{1}{3}\hat{i} + \frac{4}{3}\hat{j} - \frac{1}{3}\hat{k}$
101. In a Poisson distribution if $P[X = 3] = \frac{1}{4}P[X = 4]$ then $P[X = 5] = kP[X = 7]$ where k equals to
(a) $\frac{1}{7}$ (b) $\frac{21}{128}$ (c) $\frac{128}{21}$ (d) $\frac{21}{256}$
102. The area between the curves $y = 2 - x^2$ and $y = x^2$ is
(a)8/3 (b)4/3 (c)2/3 (d)5/3
103. A coin is tossed three times. The probabilities of getting head and tail alternatively is
(a) $\frac{1}{11}$ (b) $\frac{2}{3}$ (c) $\frac{3}{4}$ (d) $\frac{1}{4}$
104. $\int \log_{10} dx$ is
(a) $(x-1) \log_e x + c$
(b) $\log_e 10 \cdot x \log_e \left(\frac{x}{e}\right) + c$
(c) $\log_{10} e \cdot x \log_e \left(\frac{x}{e}\right) + c$
(d) $\frac{1}{x} + c$
105. If $(1+x)^n = a_0 + a_1x + a_2x^2 + \dots + a_nx^n$, then $\left(1 + \frac{a_1}{a_0}\right)\left(1 + \frac{a_2}{a_1}\right)\left(1 + \frac{a_3}{a_2}\right) + \dots \left(1 + \frac{a_n}{a_{n-1}}\right)$ is equal to
(a) $\frac{n^n}{n!}$ (b) $\frac{(n+1)^n}{n!}$ (c) $\frac{n^{n+1}}{(n+1)!}$ (d) $\frac{(n-1)^n}{n!}$
106. The value of the integral $\int_3^6 \frac{\sqrt{x}}{\sqrt{9-x} + \sqrt{x}} dx$ is
(a)1 (b)1/2 (c)3/2 (d)2
107. A survey shows that 63% of the Americans like cheese where as 76% like apples. If x% of the Americans like both cheese and apples, then we have
(a) $x \geq 39$ (b) $x \leq 63$ (c) $39 \leq x \leq 63$ (d)None
108. If $f(x) = \begin{cases} x \sin\left(\frac{1}{x}\right) & \text{for } x \neq 0 \\ 0 & \text{for } x = 0 \end{cases}$ then
(a)f is a continuous function
(b)f'(0+) exists but f'(0-) does not exist
(c)f'(0+) $\neq$ f'(0)
(d)f'(0+) and f'(0-) do not exist
109. How many proper subset of {1, 2, 3, 4, 5, 6, 7} contain the number 1 and 7 ?
(a)7 (b)31 (c)32 (d)62
110. The angle between the asymptotes of the hyperbola $27x^2 - 9y^2 = 24$ is
(a) 60° (b) 120° (c) 30° (d) 150°



111. Let $f(x) = \frac{1}{\sqrt{1-x}} + \frac{1}{\sqrt{x+3}}$

Then the domain of the real function f is

- (a) $(-\infty, -3) \cup (1, \infty)$ (b) $(1, \infty)$
(c) $(-\infty, 1) \cap (-3, \infty)$ (d) $x \neq 1, x \neq -3$

112. If $\int_{12}^{-10} f(x) dx = 6$, $\int_{100}^{-10} f(x) dx = -2$, and $\int_{100}^{-5} f(x) dx = 4$ then $\int_{-5}^{12} f(x) dx = ?$

- (a) 0 (b) -12 (c) 4 (d) -2

113. In a triangle ABC, where $A = (-4, 2)$, $B = (2, 1)$ and $C = (-4, -5)$, find $\angle ACB$.

- (a) 30° (b) 45° (c) 25° (d) 60°

114. If $\cos \theta - 4 \sin \theta = 1$, find $\sin \theta + 4 \cos \theta$

- (a) ± 1 (b) 0 (c) ± 2 (d) ± 4

115. Derivative of $\sin x^2$ with respect to $2x^2$ is:

- (a) $\frac{1}{2} \cos^2 x$ (b) $\cos\left(\frac{x^2}{2}\right)$
(c) $2 \cos x^2$ (d) $\frac{1}{2} \cos x^2$

116. $f(x) = \begin{vmatrix} x & 8 & 8 \\ 2 & x & 8 \\ 2 & 2 & x \end{vmatrix}$ has local maximum at ?

- (a) 4 (b) -4 (c) 16 (d) $\sqrt{8^2 - 2^2}$

117. Consider the matrix

$$A = \begin{vmatrix} 1 & -2 & 2 \\ 0 & 2 & 0 \\ 1 & 1 & 3 \end{vmatrix}$$

Then $A^3 = 9A$ is

- (a) an identity matrix
(b) a matrix with all entries 10
(c) a diagonal matrix
(d) None of these

118. $\sum_{k=1}^n \frac{1}{k(k+1)}$ is equal to

- (a) $\frac{1}{n}$ (b) $\frac{1}{n+1}$ (c) $\frac{n}{n+1}$ (d) $\frac{1}{n(n+1)}$

119. Find a value of $\sqrt{2 + \sqrt{2 + 2 \cos 4\theta}}$

- (a) $2 \cos 2\theta$ (b) $\cos 2\theta$
(c) $2 \cos \theta$ (d) $\sin 2\theta$

120. The value of $\sin(2 \sin^{-1}(0.6))$

- (a) 0.96 (b) $\sin(1.2)$
(c) 0.48 (d) $\sin(1.6)$

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ANSWER KEY

1.	(b)	2.	(c)	3.	(c)	4.	(c)	5.	(a)
6.	(c)	7.	(b)	8.	(d)	9.	(b)	10.	(a)
11.	(b)	12.	(d)	13.	(c)	14.	(b)	15.	(c)
16.	(c)	17.	(c)	18.	(c)	19.	(a)	20.	(d)
21.	(b)	22.	(c)	23.	(d)	24.	(d)	25.	(b)
26.	(c)	27.	(d)	28.	(b)	29.	(d)	30.	(b)
31.	(b)	32.	(b)	33.	(c)	34.	(a)	35.	(a)
36.	(b)	37.	(b)	38.	(a)	39.	(a)	40.	(a)
41.	(c)	42.	(d)	43.	(a)	44.	(c)	45.	(c)
46.	(d)	47.	(a)	48.	(b)	49.	(b)	50.	(c)
51.	(d)	52.	(b)	53.	(d)	54.	(c)	55.	(d)
56.	(d)	57.	(d)	58.	(c)	59.	(b)	60.	(a)
61.	(a)	62.	(b)	63.	(a)	64.	(d)	65.	(c)
66.	(b)	67.	(b)	68.	(c)	69.	(b)	70.	(d)
71.	(d)	72.	(b)	73.	(a)	74.	(a)	75.	(d)
76.	(a)	77.	(b)	78.	(b)	79.	(d)	80.	(b)
81.	(b)	82.	(d)	83.	(d)	84.	(d)	85.	(c)
86.	(b)	87.	(d)	88.	(c)	89.	(a)	90.	(d)
91.	(d)	92.	(b)	93.	(a)	94.	(c)	95.	(d)
96.	(b)	97.	(c)	98.	(d)	99.	(c)	100.	(c)
101.	(b)	102.	(a)	103.	(d)	104.	(c)	105.	(b)
106.	(c)	107.	(c)	108.	(a)	109.	(b)	110.	(b)
111.	(c)	112.	(B)	113.	(b)	114.	(d)	115.	(d)
116.	(b)	117.	(d)	118.	(c)	119.	(c)	120.	(a)

SOLUTION

Computer

1. (b)	2. (c)	3. (c)	4. (c)	5. (a)
6. (c)	7. (b)	8. (d)	9. (b)	10. (a)
11. (b)	12. (d)	13. (c)	14. (b)	

Reasoning

15. (b) Therefore the total number of different one way tickets:
 $= {}^7C_2 \times 2 = 42$

16. (c) What are x and y?

S	8	W	16	A	5	C	X	A	4
20	J	25	T	4	K	7	L	Y	N
J	8	W	25	T	16	A	4	K	5
	-2		+2		-4		+3		-6
									+4

S	20	L	x	A	Y	N	4
	+1		-8		+5		-10

X = 4, y = 6 Ans.

17. (c)

REASON - 5 $\Rightarrow 6 - 1 = 5$



BELIEVED - 7 $\Rightarrow 8 - 1 = 7$
 GOVERNMENT = ? Number of Alphabet - 1
 GOVERNMENT = 9

Direction Qs. 18 to 20: The cost will be minimum by using type Q thrice and s one each.

Type Q $\times 3$ + Type R $\times 1$ + Type a $\times 1 = 220$

18. (c) $140 \times 3 + 125 \times 1 + 25 \times 1 = 640$

19. (a) The number of buses required the cost is minimum is 5.

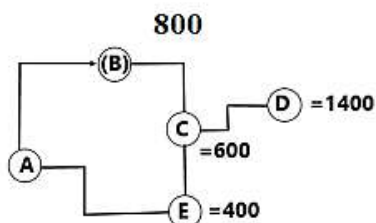
20. (d) The next lest cast is obtained by using type Q two times and type R three times the cast.

$= 2 \times 140 + 3 \times 125 = \text{Rs. } 655$

21. (b) Ankle

22. (c) $3^8 = 6561$ Ans. 8

(23-25):



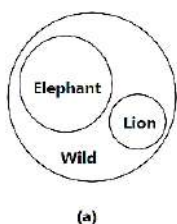
23. (d) The quantity moved A to E 2000.

24. (d) The quantity in A - B is
 $2000 - 1400 = 600$

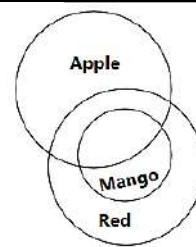
25. (b) Flow in the E-C pipeline = 1600
 $2000 - 1600 = 400$

26. (c) Woman can do the work + hours = $1 - 15$
 $\frac{18}{t} + \frac{6}{1-15} = \frac{3}{5} + \frac{8}{+}$ = the work done by a child in
 18 hours and $\frac{6}{1-15}$ = work done by a woman 6 hours
 $\frac{6}{+} + \frac{2}{1-15} = \frac{1}{5}$
 $5(8t - 90) = t^2 - 15t$
 $T^2 - 55t + 480 = 0$
 $(t - 45)(t - 10) = 0$
 $t = 45$
 $t - 15 = 45 - 15 = 30$ hours

27. (d)

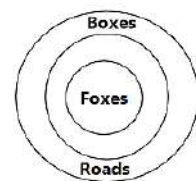


All lions are elemphtants not confirm.



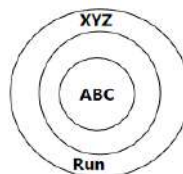
(b)

All apples are red not confirm.



(c)

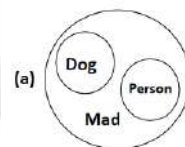
All Boxes are foxes not confirm.



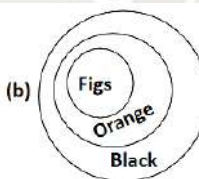
(d)

All ABC can run True. Ans. Option (d)

28. (b)

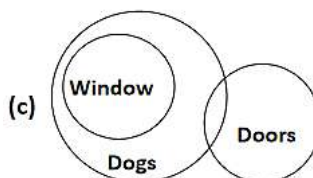


All side persons are dogs.
 Not confirm.



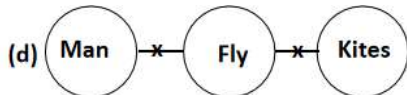
(b)

All Figs are black
 True.



(c)

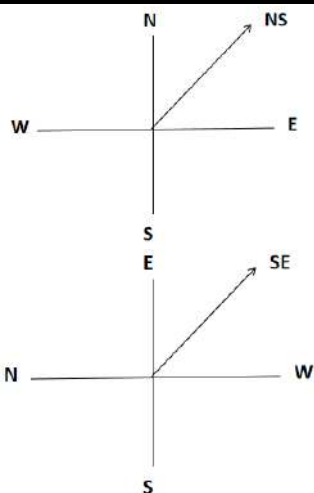
All windows are door.
 Not confirm.



(d)

All men are kites.
 Not confirm.

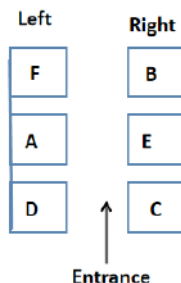
43. (a) North - East $\rightarrow$ south - East



44. (c)

Perfect squares and cubes are = 164729

Direction for Qs. 45-48



45. (c) If E sits in his office and faces the corridor C's office will be to his left.

46. (d) E's office faces A's office.

47. (a) Only A is F's neighbour.

48. (b) Directing the person to B's office.

49. (b)

Y/w	z	w/N
Green	yellow	Red

50. (c)

(a) June, July, August, September

$$2 + 3 + 3 + 2 = 10$$

(b) April, May, June, August, September, October

$$2 + 3 + 2 + 3 + 3 + 2 + 3 = 18$$

(c) April, May, June

$$2 + 3 + 2 = 7$$

(d) October November

$$3 + 2 = 5$$

Only © the number of odd is 7.

Reasoning

51.	(d)	52.	(b)	53.	(d)	54.	(c)	55.	(d)
56.	(d)	57.	(d)	58.	(c)	59.	(b)	60.	(a)
61.	(a)	62.	(b)	63.	(a)	64.	(d)	65.	(c)
66.	(b)	67.	(b)	68.	(c)	69.	(b)	70.	(d)

MATHS

71. (d) $L_1 \Rightarrow 2x + 4 - y = 0$

$$L_2 \Rightarrow 6x - 3y - 5 = 0$$

$$L_3 \Rightarrow 3(2x - y - 5/3) = 0$$

$$2x - y - 5/3 = 0$$

$$\text{Distance} = \frac{4 - (\frac{5}{3})}{\sqrt{2^2 + (-1)^2}}$$

$$= \frac{17/3}{\sqrt{5}} = \frac{17}{3\sqrt{5}} = \frac{17\sqrt{5}}{15}$$

72. (b) $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$

The equation of tangent at point $(a \cos \theta, b \sin \theta)$

$$\frac{x}{a} \cos \theta + \frac{y}{b} \sin \theta = 1$$

Given that intercept of line on the axes are equal

$$\frac{a}{\cos \theta} = \frac{b}{\sin \theta} = l$$

$$\cos \theta = \frac{a}{l} \quad \dots\dots\dots(i)$$

$$\sin \theta = \frac{b}{l} \quad \dots\dots\dots(ii)$$

$$(Eq. 1)^2 + (Eq. 2)^2$$

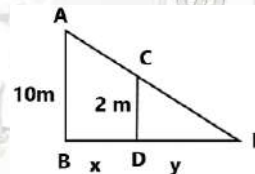
$$\sin^2 \theta + \cos^2 \theta = \frac{a^2}{l^2} + \frac{b^2}{l^2}$$

$$1 = \frac{a^2 + b^2}{l^2}$$

$$L = \sqrt{a^2 + b^2}$$

73. (a) Let x be the length between man and lamp, and y be the length of shadow.

Given



$$\frac{dx}{dt} = 2m/s$$

From the figure

$$\frac{10}{x+y} = \frac{2}{y}$$

$$8y = 2x$$

$$8 \frac{dy}{dt} = 2 \frac{dx}{dt}$$

$$\frac{dy}{dt} = \frac{1}{4} \times 2 = \frac{1}{2} m/se.$$

74. (a) $L_1 - x^2 - 2mxy - y^2 = 0$

$$L_2 - x^2 - 2nxy - y^2 = 0$$

Let L_2 is bisector of L_1

The equation of angle bisector for a pair of str. Line

represented by $ax^2 + 2hxy + by^2$

$$\frac{x^2 - y^2}{a - b} = \frac{xy}{h}$$

Angle bisector of 4 is

$$\frac{x^2 - y^2}{1 - (1)} = \frac{xy}{-m} \quad \text{Comparing to line } L_2 \text{ and eq-1.}$$

$$x^2 - y^2 = \frac{-2xy}{m}$$

$$x^2 + \frac{2xy}{m} - y^2 \quad \dots\dots\dots(1)$$

$$\frac{1}{m} = -n$$

$$mn = -1$$

$$Mn + 1 = 0$$

75. (d) $\sin^{-1} \left(\frac{2a}{1+a^2} \right) - \cos^{-1} \left(\frac{1-b^2}{1+b^2} \right) = \tan^{-1} \left(\frac{2x}{1-x^2} \right)$

Let $a = \tan \theta$, $b = \tan \phi$

LHS



$$\sin^{-1} \frac{2 \tan \theta}{1 + \tan^2 \theta} - \cos^{-1} \frac{1 - \tan^2 \theta}{1 + \tan^2 \theta}$$

$$\Rightarrow \sin^{-1} \sin^2 \theta - \cos^{-1} \cos \phi$$

$$\Rightarrow 2\theta - 2\phi$$

$$\Rightarrow 2 \tan^{-1} a - 2 \tan^{-1} b$$

$$2 \tan^{-1} \left(\frac{a-b}{1+ab} \right)$$

$$\Rightarrow 2 \tan^{-1} \left(\frac{a-b}{1+ab} \right) = \tan^{-1} \left(\frac{2x}{1-x^2} \right)$$

$$2 \tan^{-1} \left(\frac{a-b}{1+ab} \right) = 2 \tan^{-1} x$$

$$x = \frac{a-b}{1+ab}$$

76. (a) $|A| = \begin{vmatrix} 1 & 1+i+w^2 & w^2 \\ 1-i & -1 & w^2-1 \\ -i & -i+w-1 & -w^3 \end{vmatrix}$

$$R_2 \rightarrow R_2 - R_3$$

$$\Rightarrow \begin{vmatrix} 1 & 1+i+w^2 & w^2 \\ 1 & -w+i & w^2-(w^3) \\ -i & -i+w & -w^3 \end{vmatrix}$$

$$\Rightarrow \begin{vmatrix} 1 & 1+i+w^2 & w^2 \\ 1 & -w+i & w^2 \\ -i & -i+w & -w^3 \end{vmatrix}$$

$$C_1 - C_2$$

$$w^2 \begin{vmatrix} 0 & 1+i+w^2 & w^2-1 \\ 0 & -w+i & w^2-1 \\ -i+w & -i+w & -w \end{vmatrix}$$

$$w^2 (-i+w) [w-i-1-i-w^2]$$

$$w^2 (-i+w) [w-i-1-i+1+w]$$

$$w^2 (-i+w) (i-i)$$

$$w^2 (-i+w) \times 0$$

$$= 0$$

77. (b)



PB, PA is position vector of B and A.

And PC is position vector of C

C is middle point

$$\vec{PC} = \frac{\vec{PA} + \vec{PB}}{2}$$

$$2\vec{PC} = \vec{PA} + \vec{PB}$$

78. (b) a, b, c $\rightarrow$ AP

P, q, r $\rightarrow$ HP

ap, bq, cr $\rightarrow$ GP

$$2b = a + c \dots\dots\dots(1)$$

$$q = \frac{2pr}{p+r} \dots\dots\dots(2)$$

$$(bq)^2 = ap \cdot cr$$

$$\left(\frac{a+c}{2} \cdot \frac{3pr}{p+r} \right)^2 = ap \cdot cr$$

$$\left(\frac{a+c}{p+r} \right)^2 \cdot \frac{ap \cdot cr}{p^2 r^2} = \frac{ap \cdot cr}{ac}$$

$$\frac{(p+r)^2}{(a+c)^2} = \frac{ac}{ac}$$

$$\frac{(p+r)^2}{pr} = \frac{(a+c)^2}{ac}$$

$$\frac{p^2 + r^2 + 2pr}{pr} = \frac{a^2 + c^2 + 2ac}{ac}$$

$$\frac{p}{r} + \frac{r}{p} = \frac{a}{c} + \frac{c}{a}$$

79. $\vec{OA} = i + j + k$

$$\vec{OB} = 2i + 5j$$

$$\vec{OC} = 3i + 2j - 3k \quad \text{angle between } \vec{AB} \text{ \& } \vec{CD}$$

$$\vec{OC} = i - 6j - k$$

$$\vec{AB} = \vec{OB} - \vec{OA} = i + 4j - k$$

$$\vec{CD} = \vec{OD} - \vec{OC} = -2i - 8j + 2k$$

$$\text{Angle} \Rightarrow \frac{\vec{AB} \cdot \vec{CD}}{|\vec{AB}| |\vec{CD}|} = \cos \theta$$

$$\cos \theta = \frac{(-2-32-2)}{\sqrt{18} \cdot \sqrt{72}}$$

$$\cos \theta \Rightarrow \left(\frac{-36}{18\sqrt{4}} \right) = -1$$

$$\theta = \pi$$

80. (b) Value of $x^4 + 9x^3 + 35x^2 - x + 4$ for

$$x = -5 + 2\sqrt{4}$$

$$x = (-5 + 4i)$$

$$x^2 = 25 + 16i - 40i = 25 - 16 - 40i = 9 - 40i$$

$$x^4 = 81 + 1600i^2 - 720i \Rightarrow -1519 - 720i$$

$$x^3 = (9 - 40i)(-5 + 4i) = -45 + 200i + 36i - 160i^2$$

$$= -45 + 160 + 236i$$

$$= 115 + 236i$$

Put all values in equation

$$-1519 - 720i + 1035 + 2124i + 315 - 1400i$$

$$+ 5 - 4i + 4$$

$$= -160$$

81. (b) Before Rotation

$$\vec{a}_b = (2p + 1)i + j$$

Magnitude of the vector will be always same

$$|\vec{a}_0| = |\vec{a}_b|$$

$$\sqrt{(2p)^2 + 1^2} = \sqrt{(p+1)^2 + 1}$$

$$4p^2 + 1 = (p+1)^2 + 1$$

$$4p^2 = p^2 + 2p + 1$$

$$3p^2 - 2p - 1 = 0$$

$$P = 1$$

$$\text{Or } P = -1/3$$

82. (d) Average of 30 student = 45 mark

After correcting these mark were increased by 24 and 34

$$\text{The new average} = 45 + \frac{24+34}{30}$$

$$= 45 + \frac{58}{30}$$

$$= 45 + 1.93$$

$$= 46.93$$

$$= 47.$$

83. (d) $T_1 = \int_0^1 2x^2 dx, \quad T_2 = \int_0^1 2x^3 dx,$

$$T_3 = \int_1^2 2x^2 dx, \quad T_4 = \int_1^2 2x^3 dx,$$

We know that

$$0 < x < 1$$

$$\int_0^1 2x^2 \geq \int_0^1 2x^3 dx$$

$$1 < x < 2$$

$$\int_2^1 2x^3 \geq \int_2^1 2x^2 dx$$

$$T_3 > T_4$$

84. (d)

$$n = 100$$

P(E) of getting held = P

$$P(H) = P$$

$$q - P(T) = 1 - P$$

given that,

getting head of 50 coins = getting head in 51

$${}^{100}C_{50} \cdot P^{50} q^{50} = {}^{100}C_{51} \cdot P^{51} q^{49}$$

$${}^{100}C_{50} \cdot P^{50} (1-P)^{50} = {}^{100}C_{51} \cdot P^{51} (1-P)^{49}$$

$$\frac{1-P}{P} = \frac{{}^{100}C_{51}}{{}^{100}C_{50}}$$



$$\frac{1}{p} - 1 = \frac{100!}{\frac{49! 51!}{100!}} = \frac{50}{51}$$

$$\frac{1}{p} = \frac{50}{51} + 1 = \frac{101}{51}$$

$$p = \frac{51}{101}$$

85. (c) Value of the integral $\int_0^{\pi/4} \frac{\sin x + \cos x}{3 + \sin 2x}$

$$\int_0^{\pi/4} \frac{\sin x + \cos x}{3 - (\sin x)^2}$$

$$\int_0^{\pi/4} \frac{\sin x + \cos x}{3 - (1 - \sin 2x) + 1}$$

$$\int_0^{\pi/4} \frac{\sin x + \cos x}{4 - (\sin x - \cos x)^2}$$

Let $\sin x - \cos x = t$

$$(\cos x + \sin x) dx = dt$$

$$\int_{-1}^0 \frac{dt}{2^2 - t^2}$$

$$- \int_0^1 \frac{dt}{2^2 - t^2}$$

$$= \left[\frac{1}{4} \log \left| \frac{2-t}{2+t} \right| \right]_0^{-1}$$

$$\frac{1}{4} \log \left| \frac{2-(-1)}{2-1} \right| - \frac{1}{4} \log \left| \frac{2}{2} \right|$$

$$= \frac{1}{4} \log 3$$

86. (b)

India plays total 4 matches

P (getting 0) = 0.45

P (getting 1) = 0.05

P (getting 2) = 0.5

At least 7 point can be obtained in following ways

(2 2 2 1) (2 2 2 2)

$$P(E) = {}^4C_1 (P(2))^3 \cdot P(1) + {}^4C_4 (P(2))^4$$

$$= 4 \times (0.5)^3 \times 0.05 + (0.5)^4$$

$$(0.5)^3 [0.2 + 0.5]$$

$$= 0.7 \times 0.125$$

$$= 0.0875$$

87. (d) $x^2 = 4ay$

$$\Rightarrow t_1 t_2 = -1$$

Equation of tangent

$$2at_1x = 4a \left(\frac{y+at_1^2}{2} \right)$$

$$t_1x = y + at_1^2$$

$$t_1x_1 = at_1^2$$

$$x_1 = at_1$$

$$\text{similarly } x_2 = at_2$$

$$x_1x_2 = a^2 (t_1t_2)$$

$$x_1x_2 = -a^2$$

88. (c) $n(A) = 3$

$$n(B) = 4$$

$$\text{No. of one function} = 4 \times 3 \times 2$$

$$= 24$$

89. (a) The angle of intersection of the cardioids

$$r = a(1 + \cos \theta)$$

$$r = a(1 - \cos \theta)$$

$$a(1 - \cos \theta) = a(1 + \cos \theta)$$

$$2 \cos \theta = 0$$

$$\cos \theta = 0$$

$$\theta = \frac{\pi}{2}$$

90. (d)

(a) If A, B two set, then $A - B = A \cap \bar{B}$

$$A - B = A \cap (U - B)$$

Only a = only A

(b) A, B, C are set,

$$(A-B)-C = (A-C) - (B-C)$$

$$(c) \bar{A} \cup \bar{B} = \overline{A \cap B}$$

$$(d) A \cap B \cap \bar{C} \subseteq A \cap B$$

91. (d)

$$C_1 \Rightarrow x^2 + y^2 = 9$$

$$C_2 \rightarrow x^2 + y^2 - 6x - 8y + 25 = c^2$$

$$C_1(0,0), r_1 = 3$$

$$C_2(3,4) r_2 = \sqrt{3^2 + 4^2 - 25 + c^2} = \sqrt{c^2}$$

If C_2 contains the circle. C_2 then

$$C_1C_2 \leq r_2 - r_1$$

$$\sqrt{25} < \sqrt{c^2} - 3$$

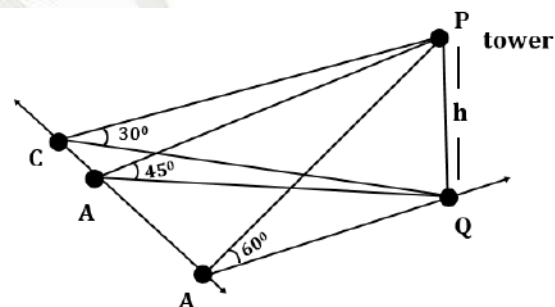
$$5 + 3 < |c|$$

$$8 < |c|, c > 8 \text{ or } c < -8$$

$$C \in (-\infty, -8) \cup (8, \infty)$$

$$C = 10$$

92. (b)



$$AQ = h \cot 60^\circ$$

$$BQ = h$$

$$CQ = \frac{h}{\sqrt{3}}$$

from right angled $\triangle ABQ$, $\angle A = 90^\circ$, we has

$$AB^2 = BQ^2 - AQ^2 = \frac{2}{3}h^2$$

$$AC^2 = CQ^2 - AQ^2 = \frac{8}{3}h^2$$

$$\frac{AC}{AB} = \sqrt{4}$$

$$AC = 2AB \text{ or } AB + BC = 2AB$$

$$AB = BC$$

$$AB \cdot BC = 1$$

93. (a)

Initial point (4,1)

(i) (1, 4) reflection about $x = y$

(ii) (3, 4) transformation through positive direction of x axis

$$(iii) (3, 4) \rightarrow 3\cos 45^\circ - 4\sin 45^\circ; 3\cos 45^\circ + 4\sin 45^\circ$$

$$\left(-\frac{1}{\sqrt{2}}, \frac{7}{\sqrt{2}} \right)$$

94. (c) R is circumradius

and $QR^2 = a^2 + b^2 + c^2$, the $\triangle ABC = ?$

$$a^2 + b^2 + c^2 = 8R^2$$

$$2R = \frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$$

$$4R^2 \sin^2 A + 4R^2 \sin^2 B + 4R^2 \cos 2C = 8R^2$$

$$\sin^2 A + \sin^2 B + \sin^2 C = 2$$

$$3 - (\cos 2A + \cos 2B + \cos 2C) = 4$$





$$\cos 2A + \cos 2B + \cos 2C = -1$$

$$-1 - 4\cos A \cos B \cos C = -1$$

$$\text{If mean } \cos A = 0 \text{ or } \cos B = 0 \text{ or } \cos C = 0.$$

$$A = 90^\circ \text{ or } B = 90^\circ \text{ or } C = 90^\circ$$

The triangle is right angle.

$$95. \Delta = \begin{vmatrix} P & b & C \\ a & q & C \\ a & b & r \end{vmatrix} = 0$$

$$R_2 \rightarrow R_2 - R_3$$

$$\begin{vmatrix} P & b & C \\ 0 & q-b & C-r \\ a & b & r \end{vmatrix} = 0$$

$$P(r(q-b) - b(c-r)) + a(b(c-r) - C(q-b)) = 0$$

$$Pr(q-b) + Pb(r-c) - ab(r-c) - ac(q-b) = 0$$

$$(Pr-ac)(q-b) + b(P-a)(r-c) = 0$$

$$\frac{Pr-ac}{(-a)(r-c)} + \frac{b}{q-b} = 0$$

$$\frac{Pr-ar+ar-ac}{(P-a)(r-c)} + \frac{b}{q-b} = 0$$

$$\frac{x(P-a)+a(r-c)}{(P-a)(r-c)} + \frac{b}{q-b} = 0$$

$$\frac{r}{r-c} + \frac{q}{p-a} + \frac{b}{q-b} = 0$$

$$\frac{p}{p-a} + \frac{q}{q-b} + \frac{r}{r-c} = \frac{p}{p-a} - \frac{q}{p-a} - \frac{b}{q-b}$$

$$\frac{p}{p-a} + \frac{q}{q-b} + \frac{r}{r-c} = \frac{p-a}{p-a} + \frac{q-b}{q-b}$$

$$\frac{p}{p-a} + \frac{q}{q-b} + \frac{r}{r-c} = 2 \text{ Ans.}$$

96. (b)

$$\sqrt{3}\cot 20^\circ - 4\cos 20^\circ = ?$$

$$\frac{\sqrt{3}\cos 20^\circ}{\sin 20^\circ} - 4\cos 20^\circ$$

$$\frac{\sqrt{3}\cos 20^\circ - 4\sin 20^\circ \cos 20^\circ}{\sin 20^\circ}$$

$$2\left[\frac{\sqrt{3}}{2}\cos 20^\circ - 2\sin 20^\circ \cos 20^\circ\right]$$

$$\frac{\sin 20^\circ}{\sin 60^\circ \cos 20^\circ - 2\sin 40^\circ}$$

$$\frac{\sin 20^\circ}{\sin 80^\circ + \sin 40^\circ - 2\sin 40^\circ}$$

$$\frac{\sin 20^\circ}{2\cos 60^\circ \sin 20^\circ} = 2 \times \frac{1}{2} = 1$$

97. (c)

$y = a \log x + bx^2 + x$ has entream value at $x = -1$ and $x = 2$, then = ?

$$f(-1) = 0$$

$$f(2) = 0$$

$$f(x) = \frac{a}{x} + 2bx + 1$$

$$f(-1) = 0, \quad f(2) = 0$$

$$-a-2b+1 = 0 \quad \frac{a}{2} + 4b = -1 \dots\dots\dots(2)$$

$$a+2b = 1 \quad \dots\dots\dots(1)$$

Solving equation (1) and (2)

$$a=2, b = -\frac{1}{2}$$

98. (d) $\vec{a}, \vec{b}, \vec{c}$ non collinear

$$\vec{a} + \vec{b} = \lambda \vec{c} \quad \vec{a} + \vec{b} + \vec{c} = ?$$

$$\vec{b} + \vec{c} = \beta \vec{a}$$

$$a + b + \vec{c} = \lambda \vec{c} + \vec{c} \Rightarrow (\lambda+1) \vec{c} \quad \dots\dots\dots(1)$$

$$a + \vec{b} + \vec{c} = \beta \vec{a} + a = (\beta+1) \vec{a} \quad \dots\dots\dots(2)$$

$$(\lambda+1) \vec{c} = (\beta+1) \vec{a}$$

But not 2 vector are collinear then

$$\lambda = -1 \text{ or } \beta = -1$$

$$\vec{a} + \vec{b} + \vec{c} = 0$$

99. (c) $a^3x + (a+1)^3y + (a+2)^3z = 0$

$$ax + (a+1)y + (a+2)z = 0$$

$$x + y + z = 0$$

has non-zero soln then.

$$D = 0 \begin{vmatrix} a^3 & (a+1)^3 & (a+2)^3 \\ a & (a+1) & (a+2) \\ 1 & 1 & 1 \end{vmatrix} = 0$$

$$a^3(a+1-a-2) - (a+1)^3(a-a-2) + (a+2)^3(a-a-1) = 0$$

$$-a^3 + 2(a+1)^3 - (a+2)^3 = 0$$

$$-a^3 + 2(a^3 + 3a^2 + 3a + 1) - (a^3 + 6a^2 + 12a + 8) = 0$$

$$2 + 6a^2 + 6a - 8 - 6a^2 - 12a = 0$$

$$-6 - 6a = 0$$

$$a = -1$$

100.(c)

$$\vec{a} = i + j$$

$$\vec{b} = j + k$$

$$\vec{c} = ?$$

$$|\vec{c}| = \sqrt{2}$$

$$\vec{c} \cdot i < 0 \quad a < 0$$

$$a^2 + b^2 + c^2 = 2$$

Satisfy these conditions

101. (b)

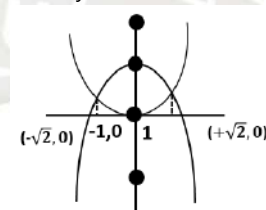
102.(a) Area between the curves

$$y = 2 - x^2 \text{ and } y = x^2 \text{ is}$$

$$x^2 = 2 - x^2$$

$$2x^2 = 2$$

$$x = \pm 1, y = 1$$



$$\text{Required area} = \int_{-\sqrt{2}}^{\sqrt{2}} (2 - x^2) - 2 \left[\int_1^{\sqrt{2}} (2 - x^2) + \int_0^1 x^2 dx \right]$$

$$\left[2x - \frac{x^3}{3} \right]_{-\sqrt{2}}^{\sqrt{2}} - 2 \left[\left[2x - \frac{x^3}{3} \right]_1^{\sqrt{2}} + \left[\frac{x^3}{3} \right]_0^1 \right]$$

$$4\sqrt{2} - \frac{4\sqrt{2}}{3} - 2 \left[2\sqrt{2} - \frac{2\sqrt{2}}{3} - 2 + \frac{1}{3} + \frac{1}{3} \right]$$

$$-\frac{4\sqrt{2}}{3} + 4\sqrt{2} - 4\sqrt{2} + \frac{4\sqrt{8}}{3} + 4 - \frac{4}{3}$$

$$4 + \frac{4}{3}(\sqrt{2})$$

$$4 - \frac{4}{3} = \frac{8}{3} \text{ sq unit option a}$$

103.(d)

$$n(s) = 2^3 = 8$$

$$n(E) = (HTH), (T, H, T)$$

$$= 2$$

$$P(E) = \frac{n(E)}{n(S)} = \frac{2}{8} = \frac{1}{4}$$

104.(c)

$$\int \log_{10} x dx$$



$$\int \frac{\log_e x}{\log_e 10} dx$$

$$\log_{10} e \int I_d \log_e x dx$$

$$\log_{10} e \cdot \left[\log_e x \cdot x - \int \left(x \cdot \frac{1}{x} \right) dx \right]$$

$$\log_{10} e [x \log_e x - x] + c$$

$$\log_{10} e [x(\log_e x - \log_e e)]$$

$$\log_{10} e \cdot x \log_e \left(\frac{x}{e} \right) + c$$

105.(b) $(1+x)^n = a_0 + a_1x + a_2x^2 + \dots + a_nx^n$ then

let take the value of $n = 2$

$$(1+x)^2 = 1 + 2x + x^2$$

$$a_2 = 1, a_1 = 2, a_0 = 1$$

$$\text{value} = \left(1 + \frac{a_1}{a_0}\right) \left(1 + \frac{a_2}{a_1}\right) \left(1 + \frac{a_3}{a_2}\right)$$

$$= \left(1 + \frac{2}{1}\right) \left(1 + \frac{1}{2}\right) \left(1 + \frac{0}{1}\right) \dots$$

$$= \left(\frac{3}{2}\right) (3) (1)$$

$$= \frac{9}{2}$$

Put the value of $n = 2$ in option.

$$= \frac{(n+1)^n}{n!} \text{ satisfy.}$$

106. (c)

$$I = \int_3^6 \frac{\sqrt{x}}{\sqrt{9-x} + \sqrt{x}} dx \dots\dots\dots(1)$$

$$\text{Prop } \int_a^b \frac{f(x)}{f(x)+g(x)} = \int_a^b \frac{f(a+b-x)}{f(a+b-x)+g(a+b-x)}$$

$$I = \int_3^6 \frac{\sqrt{9-x}}{\sqrt{9-x} + \sqrt{9-x}} dx$$

$$I = \int_3^6 \frac{\sqrt{9-x}}{f(x)+\sqrt{9-x}} dx \dots\dots\dots(2)$$

Equation 1 + 2

$$2I = \int_3^6 \frac{\sqrt{x} + \sqrt{9-x}}{\sqrt{x} + \sqrt{9-x}} dx$$

$$2I = \int_3^6 1 \cdot dx$$

$$\frac{1}{2} [x]_3^6 = \frac{6-3}{2} = \frac{3}{2}$$

107.(c)

$n(A)$ = no. of person like cheese = 63%

$n(B)$ = no. of person like apple = 76%

$n(A \cap B)$ = like both = $x\%$

let no. of people = 100

$$n(A) = 63, n(B) = 76, n(A \cap B) = n$$

$$n(A) + n(B) - n(A \cup B) \leq n(A \cap B) \leq \min[n(A), n(B)]$$

$$63 + 76 - 100 \leq n(A \cap B) \leq 63$$

$$39 \leq n \leq 63$$

108.(a)

$$f(x) = \begin{cases} x \sin(1/x) & x \neq 0 \\ 0 & x = 0 \end{cases}$$

LHL

$$\lim_{x \rightarrow 0^-} x \sin\left(\frac{1}{x}\right)$$

$$= 0 \times \sin\left(\frac{1}{0}\right)$$

$$= 0 \times [\text{oscillating fun } [-1, 1]]$$

$$= 0$$

RHL

$$\lim_{x \rightarrow 0^+} x \sin\left(\frac{1}{x}\right)$$

$$= 0 \times \text{oscillating fun}$$

$$= 0$$

$$f(0) = 0 \quad \text{LHL} = \text{RHL} = f(0)$$

$f(x)$ is continuous.

109.(b) Proper subset of $\{1, 2, \dots, 7\}$ contain the no. 1 and 7.

$$\text{proper} = \{ {}^5C_0 + {}^5C_1 + {}^5C_2 \dots {}^5C_4 + {}^5C_5 \} - 1$$

$$\text{subse} \Rightarrow 2^5 - 1 = 31$$

110.(b) The angle between asymptotes of the hyperbola $27x^2 - 9y^2 = 24$

$$\frac{x^2}{24/27} - \frac{y^2}{24/9} = 1 \quad a = \sqrt{\frac{24}{27}}$$

$$b = \sqrt{\frac{24}{9}}$$

Angle between asymptotes = $2\tan^{-1} b/a$

$$= 2\tan^{-1} \left(\frac{\sqrt{24/3}}{\sqrt{24/3}\sqrt{3}} \right)$$

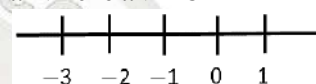
$$= 2\tan^{-1}(\sqrt{3})$$

$$= 2 \times \frac{\pi}{3} \Rightarrow \frac{2\pi}{3} \Rightarrow 120^\circ$$

111. (c) domain of function $f(x) = \frac{1}{\sqrt{1-x}} + \frac{1}{\sqrt{x+3}}$

$$1-x > 0 \text{ and } x+3 > 0$$

$$x < 1 \text{ and } x > -3$$



$$x \in (-3, 1)$$

$$x \in (-\infty, 1) \cap (-3, \infty)$$

112. (b)

$$\int_{100}^{-5} f(x) dx = 4, \int_{100}^{-10} f(x) dx = -2$$

$$\int_{12}^{-10} f(x) dx = 6,$$

$$[f'(x)]_{100}^{-5} = 4f'(-5) - f'(100) = 4$$

$$[f'(x)]_{100}^{-10} = -2f'(-10) - f'(100) = -2$$

$$[f'(x)]_{190}^{10} = 6f'(-10) - f'(120) = 6$$

$$\int_{-5}^{12} f(x) dx = ? \quad f'(-5) - f'(-10) = 6$$

$$= [f'(x)]_{-5}^{12} \quad f'(12) = f'(-10) - 6$$

$$= f'(-10) - 6 - f'(-5)$$

$$f'(-50) - f'(-5) - 6$$

$$-6 - 6$$

$$= -12$$

113. (b)

$$A(-4, 2)$$

$$B(2, 1)$$

$$C(-4, -5)$$

$$\angle ACB = ?$$

$$a^2 = \sqrt{(2 - (-4))^2 + (6)^2} = \sqrt{72} = 6\sqrt{2}$$

$$a^2 = 72$$

$$b^2 = (-4 - (-4))^2 + (2 + 5)^2 = 0 + 49 = 49$$

$$c^2 = (2 + 4)^2 + (2 - 1)^2 = 36 + 1 = 37$$

$$\cos c = \frac{a^2 + b^2 - c^2}{2ab} = \frac{72 + 49 - 37}{2 \times 6\sqrt{2} \times 7}$$

$$\cos c = \frac{72 + 12}{14 \times 6 \times \sqrt{2}} = \frac{84}{14 \times 6} = \frac{1}{\sqrt{2}}$$

$$\cos c = \frac{1}{\sqrt{2}}, \quad C = 45^\circ$$

114. (d)

$$\cos \theta - 4 \sin \theta = 1 \quad \sin \theta + 4 \sin \theta = ?$$

$$(\cos \theta - 4 \sin^2 \theta)^2 = 1$$

$$\cos \theta + 16 \sin^2 \theta = 8 \cos \theta \cdot \sin \theta = 1$$

$$(\sin \theta + 4 \cos \theta)^2 = \sin^2 \theta + 16 \cos^2 \theta + 8 \cos \theta \cdot \sin \theta$$

$$= \sin^2 \theta + 16 \cos^2 \theta + \cos^2 \theta + 16 \sin^2 \theta - 1$$

$$(\sin \theta + 4 \cos \theta)^2 = 1 + 16 - 1 \Rightarrow 16$$

$$(\sin \theta + 4 \cos \theta) = \pm 4$$



115. (d)

Derivative of $\sin x^2$ with respect to $2x^2$

$$u = \sin x^2, v = 2x^2$$

$$\frac{du}{dx} = \cos x^2 \cdot 2x \cdot \frac{dv}{dx} = 4x$$

$$\frac{du}{dv} = \frac{du/dx}{dv/dx} = \frac{\cos x^2 \cdot 2x}{4x}$$

$$\frac{du}{dv} = \frac{1}{2} \cos x^2$$

116. (b)

$$f(x) = \begin{vmatrix} x & 8 & 8 \\ 2 & x & 8 \\ 2 & 2 & x \end{vmatrix} \text{ has local maximum at}$$

$$f(x) = \begin{vmatrix} x-8 & 0 & 8 \\ 2-x & x-8 & 8 \\ 0 & 2-x & x \end{vmatrix}$$

$$= (x-8)[x(x-8) - 8(2-x)] + 8(2-x)^2$$

$$= (x-8)[x^2 - 8x - 16 + 8x] + 8(x-2)^2$$

$$f(x) = (x-8)(x^2 - 16) + 8(x-2)^2$$

$$= x^3 - 16x - 8x^2 + 128 + 8(x^2 + 4 - 4x)$$

$$f^9(x) = x^3 - 16x - 8x^2 + 128 + 8x^2 - 32x + 32$$

$$f'(x) = 3x^2 - 48$$

117. (d)

$$A = \begin{bmatrix} 1 & -2 & 2 \\ 0 & 2 & 0 \\ 1 & 1 & 3 \end{bmatrix}$$

$$A^3 = A^2 A$$

$$= \begin{bmatrix} 3 & -4 & 8 \\ 0 & 4 & 0 \\ 4 & 3 & 11 \end{bmatrix} \begin{bmatrix} 1 & -2 & 2 \\ 0 & 2 & 0 \\ 1 & 1 & 1 \end{bmatrix}$$

$$A^3 = \begin{bmatrix} 11 & -6 & 30 \\ 0 & 8 & 0 \\ 15 & 9 & 41 \end{bmatrix}$$

$$A^3 - 9A$$

$$\begin{bmatrix} 2 & 12 & 12 \\ 0 & -8 & 0 \\ 6 & 0 & 16 \end{bmatrix}$$

118. (c)

$$\sum_{K=1}^n \frac{1}{K(K+1)} = ?$$

$$\sum_{K=1}^n \frac{(K+1)-K}{K(K+1)} = ?$$

$$\sum_{K=1}^n \frac{1}{K} - \frac{1}{K+1}$$

$$1 - \frac{1}{2} + \frac{1}{2} - \frac{1}{3} + \frac{1}{3} - \frac{1}{4} \dots \frac{1}{n} - \frac{1}{n+1}$$

$$= 1 - \frac{1}{n+1}$$

$$= \frac{n+1-1}{n+1} = \frac{n}{n+1}$$

119. (c)

Find a value of $\sqrt{2 + \sqrt{2 + 2 \cos 4\theta}}$

$$= \sqrt{2 + \sqrt{2 + 2(2\cos^2\theta - 1)}}$$

$$= \sqrt{2 + \sqrt{2 + 4\cos^2\theta - 2}}$$

$$= \sqrt{2 + 2\cos 2\theta}$$

$$= \sqrt{2 + 2(2\cos^2\theta - 1)}$$

$$= \sqrt{4\cos^2\theta}$$

$$= 2|\cos \theta|$$

120. (a) The value of $\sin(2 \sin^{-1}(0.6))$

$$2 \sin^{-1} x = \sin^{-1}(2x \cdot \sqrt{1-x^2})$$

$$\sin(2 \sin^{-1}(0.6))$$

$$\Rightarrow \sin(\sin^{-1}(2 \times 0.6 \cdot \sqrt{1-(0.6)^2}))$$

$$= 2 \times 0.6 \times \sqrt{1-0.36}$$

$$= 1.2 \times 0.8$$

$$= 0.96$$



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